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BCACACS 203

Second Semester B.C.A. Degree Examination, June/July 2026
(SEP) (2024-25 Batch Onwards)
COMPUTATIONAL MATHEMATICS

Time : 3 Hours

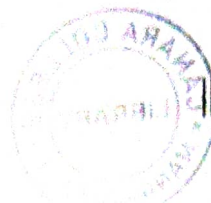
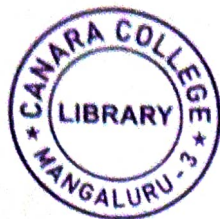
Max. Marks : 80

- Note :** 1) Answer **any ten** questions from Part – A and **one full** question from **each** Unit in Part – B.
2) Scientific calculators are **allowed**.

PART – A**(10×2=20)**

1. a) What are the two types of errors ?
- b) Find the relative error of the number 8.6 if both of its digits are correct.
Given $E_A = 0.05$.
- c) If Δ is the forward difference operator then $\Delta^3 Y_3 = ?$
- d) What is a power function ?
- e) What is Exponential function ?
- f) Write the Simpson's 3/8 rule for $\int_{x_0}^{x_n} y dx$.
- g) If $A = \begin{bmatrix} 0 & 2 & 3 \\ -2 & 0 & 5 \\ -3 & -5 & 0 \end{bmatrix}$ show that $A' = -A$.
- h) Define square matrix. Give example.
 - i) Define symmetric matrix. Give example.
 - j) Write Runge-Kutta 4th order formula.
 - k) Write Adams-Moulton predictor-corrector formula.
 - l) Write Milne's Predictor formula.

P.T.O.

**PART – B****Unit – I**

2. a) Find the real root of the following equation : $f(x) = x^3 - x - 1 = 0$ using Bisection method. (Carry out 3 iterations)
- b) Certain corresponding values of x and $\log_{10} x$ are (300, 2.4771), (304, 2.4829), (305, 2.4843) and (307, 2.4871). Find $\log_{10} 301$ using Lagrange's Interpolation Formula.
- c) From the following table, find the number of students who obtained less than 45 marks using Newton's forward difference interpolation formula. **(5+5+5)**

Marks	30 – 40	40 – 50	50 – 60	60 – 70	70 – 80
No. of Students	31	42	51	35	31

3. a) Use the method of Iteration to find a positive root, between 0 and 1 of the equation $xe^x = 1$. (Carry out 3 iterations)
- b) In the table below the values of y are consecutive terms of a series of which the number 21.6 is the 6th term. Find the first term of the series using Newton's forward difference interpolation formula.

X	3	4	5	6	7	8	9
Y	2.7	6.4	12.5	21.6	34.3	51.2	72.9

- c) Using the following table find $f(x)$ as a polynomial in x using Newton's general interpolation formula. **(5+5+5)**

x	-1	0	3	6	7
f(x)	3	-6	39	822	1611

Unit – II

4. a) Certain experimental values of x and y are given below :

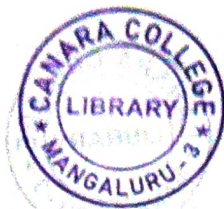
x	0	2	5	7
y	-1	5	12	20

If $y = a_0 + a_1x$, find approximate values of a_0 and a_1 .

- b) Calculate the first and second derivatives of the function tabulated below at the point $X = 2.2$ using Newton's backward difference formula.

X	1.0	1.2	1.4	1.6	1.8	2.0	2.2
Y	2.7183	3.3201	4.0552	4.9530	6.0496	7.3891	9.0250

- c) Evaluate $I = \int_1^3 \frac{1}{x} dx$ by Simpson's 1/3 rule with 4 strips. **(5+5+5)**



5. a) From the following table of values of x and y , obtain $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ for $x = 6$ using Newton's backward difference formula.

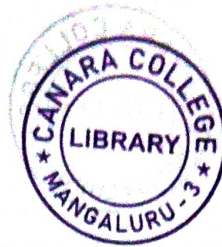
x	0	1	2	3	4	5	6
y	6.9897	7.4036	7.7815	8.1291	8.4510	8.7506	9.0309

- b) Evaluate $I = \int_0^1 \frac{1}{(1+x)} dx$ by trapezoidal rule when $h = 0.25$, correct to three decimal places.
- c) Determine the constants a and b by the method of least squares such that $Y = ae^{bx}$ fit the following data. (5+5+5)

x	2	4	6	8	10
y	4.077	11.084	30.128	81.897	222.62

Unit – III

6. a) Compute the Inverse of the matrix $A = \begin{bmatrix} 2 & 4 & 3 \\ 0 & 1 & 1 \\ 2 & 2 & -1 \end{bmatrix}$.
- b) Solve the following system using Gauss-Seidal method. Carry out 3 iterations.
- $$\begin{aligned} 10x + 2y + z &= 9 \\ 2x + 20y - 2z &= -44 \\ -2x + 3y + 10z &= 22 \end{aligned}$$
- c) Solve the following system using Gauss elimination method.
- $$\begin{aligned} 5x - 2y + z &= 4 \\ 7x + y - 5z &= 8 \\ 3x + 7y + 4z &= 10 \end{aligned} \quad (5+5+5)$$
7. a) Solve the following system using Jacobi's method. Carry out 3 iterations.
- $$\begin{aligned} 83x + 11y - 4z &= 95 \\ 7x + 52y + 13z &= 104 \\ 3x + 8y + 29z &= 71 \end{aligned}$$



b) $A = \begin{bmatrix} 2 & 3 & 2 \\ 1 & 0 & 0 \\ 2 & 0 & 2 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 2 & 1 \\ 2 & 3 & 4 \\ 1 & -2 & 3 \end{bmatrix}$ form AB and BA .

c) For the matrix $A = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$ compute A^2 and A^3 .

(5+5+5)

Unit – IV

8. a) Given $dy/dx - 1 = xy$ and $y(0) = 1$, Using Taylor's series compute $y(0.1)$, correct to four decimal places.
- b) Determine the value of y when $x = 0.1$ given that $y(0) = 1$ and $y' = x^2 + y$, $h = 0.05$, using Euler's modified method.
- c) Tabulate the solution of $dy/dx = x + y$, $y(0) = 0$ when $x = 0$ and $h = 0.1$. Find $y(0.4)$ using Milne's formula. Given $y(0.1) = 0.0052$, $y(0.2) = 0.0214$ and $y(0.3) = 0.0499$.
- (5+5+5)
9. a) Solve by Euler's method, the equation $dy/dx = x + y$, $y(0) = 0$. Choose $h = 0.2$ and compute $y(0.4)$.
- b) Given $dy/dx = y - x$ where $y(0) = 2$ and $h = 0.1$. Find $y(0.1)$ using Runge-Kutta fourth order formula.
- c) Using Adams-Moulton formula, solve $y' = 1 + y^2$ where $y = 0$ when $x = 0$ and $h = 0.2$. Find $y(0.8)$. Given that $y(0.2) = 0.2027$, $y(0.4) = 0.4228$, $y(0.6) = 0.6841$.
- (5+5+5)